Disadvantages

1. Assembling and repairing of parts is difficult.
2. It needs special lubricating oil.
3. It has very less ground clearance.
4. High cost and heavy weight.
5. It is suitable only for heavy vehicles.

4.11 Differential housing

Rear axle casing or housing is the one which has the differential unit and the half shaft within it. It helps in preventing formation of dust particles inside the axles. At the same time it provides the way for supplying the lubricating oil. The differential housing is generally classified in to two different types. They are,

1. Banjo type
2. Split type

4.11.1 Banjo type

As the complete structure of this casing looks like a banjo musical instrument, it is named as banjo type. Vacuum pipes are connected at both ends of it. At the middle the differential unit is placed. This type is generally used in all types of vehicles i.e. light and heavy vehicles. Figure 4.11.1 shows the view of the Banjo type differential housing

Advantages

1. Less weight
2. It withstands more weight.
3. Any repairs in this system can be done without dismantling in rear axle.

4.11.2 Split type

In this type, two vacuum tubes are connected with both sides of the differential casing by using bolts. In this type, if any fault occurs, the parts can be removed separately and repaired. Hence this type of housing is used in very few vehicles only. Figure 4.11.2 shows the split type differential housing

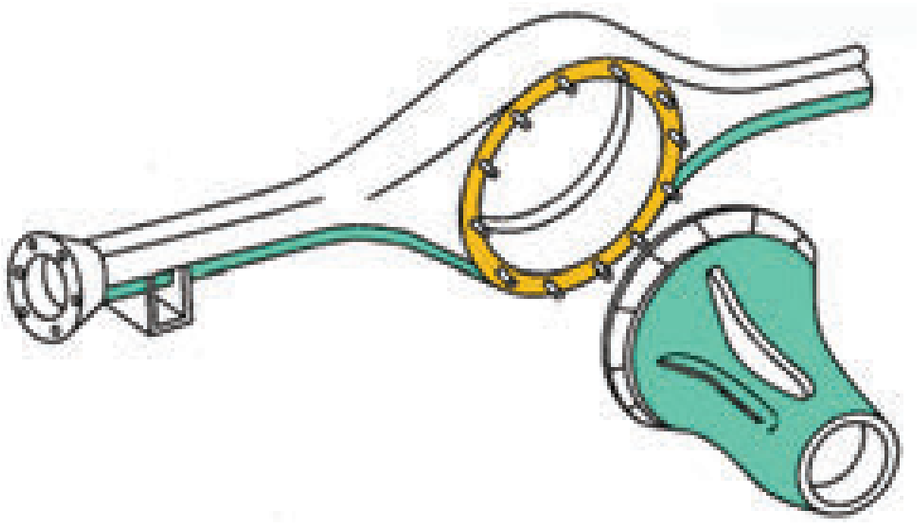


Figure 4.11.1 Banjo type differential housing

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